

Ruaridh Murdoch

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I'm a third-year Design Engineering MEng student at Imperial College London. I take products from early concept through CAD, electronics, prototyping and testing, usually with a user's needs as the starting point. I'm most interested in physical products people use every day, where details like how a mechanism feels or how long it lasts decide whether people enjoy using them. I'm methodical, open to other people's ideas and comfortable working across engineering and design teams, and I'm always looking for the next thing to build.

Education

MEng Design Engineering, Imperial College London

10/2024 – 06/2028 | London, UK

- Head of School Prize (Year 2): ranked 1st in the Design Engineering cohort
- First class in both years: 77.9% (Year 2), 72.7% (Year 1)

Projects

SwivelCut: Precision Cutting Tool for D&T Classrooms, Group of 4

2026

- Conducted cross-school user research (11 interviews, 92-person survey), finding that D&T students' manual cutting tools couldn't accurately reproduce their own designs.
- Led CAD, CMF and prototyping of a classroom cutting arm with swappable tracing and cutting heads, with a sprung blade mount cutting peak force from 28 N to 8 N.
- Validated with 18 students across Years 7 - 12, costed at £299 retail, and presented at the Great Exhibition Road Festival, drawing investor interest.

Teach-by-Touch Robotic Arm, Group of 2

2025

- Co-built a 4-joint robotic arm that learns by being moved by hand: record a sequence, insert a cartridge, press play to repeat it at the original speed.
- Developed low-cost joints from scratch using DC motors, 3D-printed gear reductions and potentiometer feedback, driven by a four-state finite state machine.

Rhythm Rush: Social Play System for Queues, Group of 5

2025

- Co-ran 21 street interviews and observations, finding that in an individualistic society people want to connect but fear intruding unless they share a context, making event queues an untapped social space.
- Co-designed a 4-player motion-tracked dance game that lets queueers play, watch or walk past, then facilitated live queue testing to add a dance guide and creativity-based scoring for hesitant users.

Experience

Undergraduate Teaching Assistant, Imperial College London

10/2025 – Present | London, UK

- Teach over 90 students a week in Physical Computing labs on the Bela embedded platform, working with Prof. Andrew McPherson, its creator.
- Previously mentored over 90 first-years weekly across maths, mechanics, materials and computing, adapting explanations to each student.

Work Experience, Paddington Robotics

06/2026 | London, UK

- Soldered and assembled electronics and hardware for an ML-driven trolley that follows shop workers while they restock.
- Worked alongside the engineering team, learning their human-centred approach of building robots that support workers rather than replace them.

Skills

Electronics & Embedded: ESP32, Arduino (C++), MicroPython, soldering, circuit prototyping, PID and state-machine control

Mechanical & CAD: Fusion 360, AutoCAD, gear trains, belt drives, tolerancing, DFM / DFA, technical drawing

Simulation & Analysis: Ansys Mechanical (FEA), Fluent (CFD), Discovery, MATLAB, Granta EduPack

Data & Machine Learning: pandas, scikit-learn, SVR, random forests, logistic regression, feature engineering

Design & Research: User interviews, surveys, stakeholder mapping, benchmarking, LCA, Figma, InDesign, Photoshop

Interests

Climbing: NIBAS/NICAS lead climber, ex-Glasgow Youth Squad, Imperial Mountaineering Club, Lakeland Mountain Marathon

Game Development: Two-time Duck Sauce Game Jam winner, shipping original Unity / C# games in 48 hours

Sailing & Water Sports: RYA Day Skipper, Scottish coastal passages, kayaking, coasteering, open-water swimming